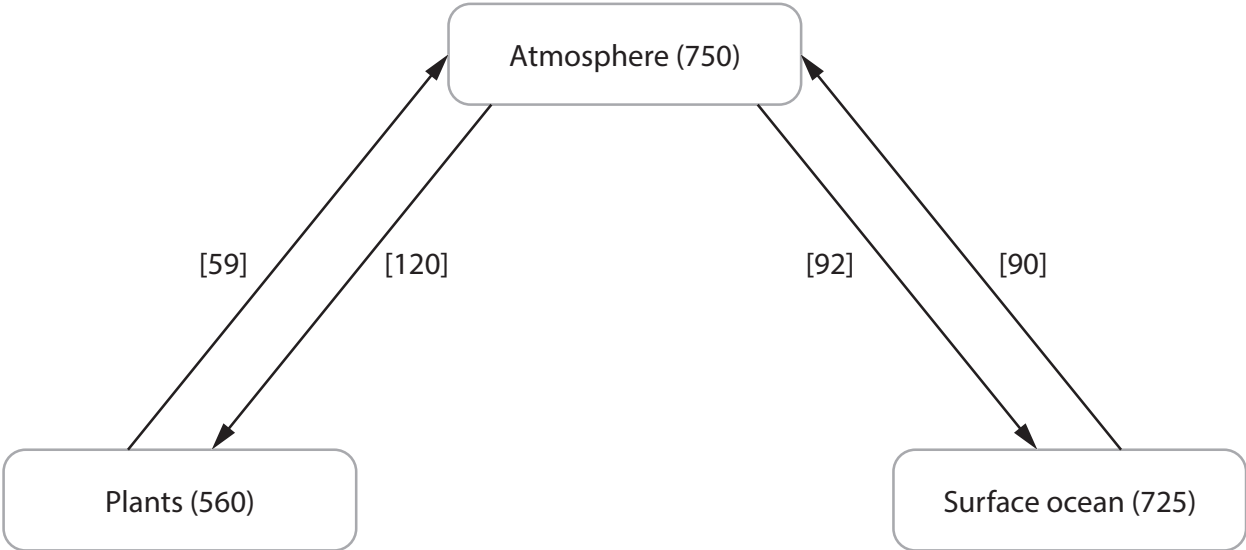


6 The diagram shows part of the carbon cycle.



Key

The numbers in smooth brackets () show the mass of carbon in petagrams found in either the atmosphere, plants or surface ocean.

The numbers in square brackets [] show the mass of carbon in petagrams transferred between the atmosphere and the plants or surface ocean per year.

A petagram is equal to 10^{15} grams.

(a) Carbon is found in carbohydrates.

The table shows some carbohydrates that may be found in plants and animals.

For each carbohydrate, put **one** cross ☒ in the appropriate box, in each row, to show where these carbohydrates are produced.

(4)

Carbohydrate	Carbohydrate produced by			
	both plants and animals	plants but not animals	animals but not plants	neither plants nor animals
Amylose	☒	☒	☒	☒
Glucose	☒	☒	☒	☒
Glycogen	☒	☒	☒	☒
Starch	☒	☒	☒	☒

(b) The mass of carbon recycling between the atmosphere and the plants, and the atmosphere and the oceans, are roughly balanced.

(i) State how carbon is exchanged between the atmosphere and the oceans.

(1)

(ii) Calculate the increase in the mass of carbon in plants as a result of carbon exchange between the plants and the atmosphere in one year.

Give your answer in kilograms.

(1)

Answer.....kg

(iii) Describe how the carbon present in sugars in the plants is returned to the atmosphere in the carbon cycle.

(3)

(c) Anthropogenic climate change is a result of an increase in the mass of carbon in the atmosphere.

(i) State what is meant by the term **anthropogenic climate change**.

(2)

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(ii) Explain **one** way in which the effects of anthropogenic climate change can be reduced.

(2)

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(Total for Question 6 = 13 marks)